r/aiwars

AI expected to account for over half of data center power consumption by the end of 2025

Posted by u/lovestruck90210 • • 0 points • 5 comments

Despite the constant gaslighting from AI bros, the impact of AI on power consumption and the environment doesn't seem to be all that great. The Guardian [reports](https://www.theguardian.com/environment/2025/may/22/ai-data-centre-power-consumption):

> Artificial intelligence systems could account for nearly half of datacentre power consumption by the end of this year, analysis has revealed. The estimates by Alex de Vries-Gao, the founder of the Digiconomist tech sustainability website, came as the International Energy Agency forecast that AI would require almost as much energy by the end of this decade as Japan uses today. De Vries-Gao's calculations, to be published in the sustainable energy journal Joule, are based on the power consumed by chips made by Nvidia and Advanced Micro Devices that are used to train and operate AI models. The paper also takes into account the energy consumption of chips used by other companies, such as Broadcom. The IEA estimates that all data centres – excluding mining for cryptocurrencies – consumed 415 terawatt hours (TWh) of electricity last year. De Vries-Gao argues in his research that AI could already account for 20% of that total.

And:

> By the end of 2025, De Vries-Gao estimates, energy consumption by AI systems could approach up to 49% of total datacentre power consumption, again excluding crypto mining. AI consumption could reach 23 gigawatts (GW), the research estimates, twice the total energy consumption of the Netherlands.

Okay, now I know what you're going to say. "This is all anti/luddite propoganda! We'll have magical technologies that will improve AI efficiency tremendously and this'll all be a big nothing burger and we can get to generating our anime girl 'content' without feeling bad about it!". The problem with this line of reasoning is something called ["Jevon's Paradox"](https://news.northeastern.edu/2025/02/07/jevons-paradox-ai-future/). The idea is that while technological advancements make using a given resource more efficient, demand for that resource also increases. This leads to larger overall consumption of that resource. Jevon wrote about this in 1860's England, when noticed that as steam engines became more fuel-efficient, coal consumption increased rather than decreased. This happened because cheaper, more efficient energy made it more accessible for more factories and industrial activities, thereby increasing demand for coal overall.

I expect this post be met with some mix of:
* Strange diatribes about how your local model doesn't consume a lot of power like if that's what is even being discussed here.
* The regurgitation of those little infographics with no sources as if they should be treated with anything other than ridicule and dismissal.
* Links to dubious, unscientific Substack articles.
* Weird, angry rants about perceived hyprocisy, "why don't you go do REAL activism" and other such nonsense.

But hey, let's see.

Comments

u/KamikazeArchon • 2 points • 2025-06-01 01:13 UTC

>Despite the constant gaslighting from AI bros, the impact of AI on power consumption and the environment doesn't seem to be all that great.

The entirety of *all* datacenter power consumption in the world today accounts for only about 2% of human power consumption.

Further, Jevon's Paradox is something that *can* happen but certainly is not *necessary* or even common. We have actual current trendlines to look at. Even with total datacenter energy use rising, and even with AI-driven demand factored in, global *emissions* from data centers - the actual thing that is harmful - are expected to *fall* in 2024-2026.

Source: [International Energy Agency](https://iea.blob.core.windows.net/assets/6b2fd954-2017-408e-bf08-952fdd62118a/Electricity2024-Analysisandforecastto2026.pdf) report, particularly p61.

u/klc81 • 2 points • 2025-05-31 22:54 UTC

https://preview.redd.it/lz0zuyg7674f1.png?width=461&format=png&auto=webp&s=fe6d7ba4f1fa1a5ee2569037e67690487d70952e

u/Beautiful-Lack-2573 • 2 points • 2025-05-31 22:01 UTC

Here's ChatGPT ("man's best friend"), running in a clean account to eliminate memory biases, summarizing critiques of this article. TL;DR - shoddy modeling and lack of understanding of technology.

Take it away, ChatGPT:

De Vries-Gao’s May 2025 paper dramatically overstates AI’s energy use by equating “GPU power specs” with actual generative-AI consumption. Here are the main flaws:

1. Using Max TDP as Reality He multiplies GPU shipments by each card’s thermal-design-point (TDP) and assumes 100 percent utilization for AI. In truth, large-scale training throttles power (mixed

precision, variable batch sizes), and inference often runs at 40–60 percent of TDP—or far less—so treating every GPU as a constant full-load device inflates energy forecasts.
2. Ignoring the Training vs. Inference Gap His model doesn't distinguish training from inference. Real-world inference is 10×–100× more efficient (lower power per token) than training, and many data centers now use dedicated inference hardware or downclock GPUs to \~200–300 W. Blurring these profiles means his numbers assume every GPU is doing heavy, full-power training all the time.
3. Overlooking Idle/Mixed Use De Vries-Gao's scenario assumes every datacenter GPU is constantly crunching AI workloads. In practice, GPUs hibernate between jobs, downclock when idle, or run non-AI tasks. If average utilization is only 50 percent (or less), that alone cuts his energy estimate roughly in half.
4. Conflating Manufacturing with Operation He sometimes treats the energy cost of fabricating GPUs as if it's all annual operational energy. Fabrication energy is amortized over 3–5 years of use; counting it entirely in one year balloons the AI footprint beyond reality.
5. Supply vs. Actual Demand Pointing to GPU supply bottlenecks and raw power ratings doesn't mean companies will—or can—run every GPU at full tilt. Deployments are governed by total cost of ownership, ROI, carbon targets, and actual workload needs. Rightsizing clusters for specific tasks inherently limits power draw.
6. Empirical Data Contradicts His Claims Recent measurements show North American data centers doubled from \~2.7 GW to \~5.3 GW (end 2022 to end 2023), but that includes all servers (GPUs, CPUs, storage, cooling). Peak GPU racks rarely hit max-rated power. Moreover, a typical generative-AI query uses \~0.0003 kWh—only \~5× a standard web search—because GPUs multiplex thousands of inference requests per second, keeping average draw far below "number of GPUs × TDP."

Bottom Line: **By treating a GPU's maximum power rating as equivalent to continuous generative-AI load—ignoring efficiency gains, idle periods, mixed use, and amortized manufacturing costs—De Vries-Gao's May 2025 article inflates AI's real energy footprint.**

u/Beautiful-Lack-2573 • 6 points • 2025-05-31 21:50 UTC

Not this guy again.

The author of the paper has published a lot more, and he has a track record for being uniformly criticized for his inaccurate assumptions. Experts and policy makers do not take this person seriously. He's an environmental activist first, anti-AI activist second, and researcher a distant third. Disappointed that The Guardian is being so credulous.

>are based on the power consumed by chips made by Nvidia and Advanced Micro Devices that are used to train and operate AI models.

And here, as numerous people have again pointed out over the past week, is the rub. These chips are used for EVERYTHING. Pretty much any workloads that is able to be parallelized in any way is moving to be GPU-accelerated, including video encoding and decoding, rendering, social media algorithms. Equating "GPUs" with "AI" is either intentional deceit or a massive blunder. (Equating ALL OF AI with GENERATIVE AI would be a second massive blunder, but he doesn't even get to that.)

Here's Physics Today, not a dubious site, on one his past studies:
[https://pubs.aip.org/physicstoday/article/77/4/28/3278527]
(https://pubs.aip.org/physicstoday/article/77/4/28/3278527)

As always, these papers have the misleading "consumes as much power as country X in a year and as much as country Y in a month" comparisons, which is a big red flag. It's not about putting serious numbers in front of policy makers, it's about impressing the reader with shocking numbers.

That AI "could" consume as much power as Bitcoin by the end of the year. Again, trying to sound impressive, just making an own goal. So AI - a transformative technology used by a BILLION people - is still using LESS power than freakin' Bitcoin, the biggest waste of power ever?

Ok, enough tearing into this guy.

Of course power consumption matters in the big picture. No one is saying AI power consumption won't dramatically increase. It'll continue to increase steeply, just as data center power consumption was ALWAYS going to increase massively, but AI has steepened the curve, basically from a doubling every five years to a doubling every three or four.

Yes, that's right. There is no scenario where data center power - with or without AI - doesn't increase tenfold over the coming decade, and another tenfold over the decade after. This is happening whether or not anyone uses ChatGPT.

u/NuOfBelthasar • 0 points • 2025-05-31 21:37 UTC

>I expect this post be met with some mix of: \[...\]

How about a link to the [white paper](https://www.cell.com/joule/fulltext/S2542-4351(25)00142-4) so you can judge for yourself the methods and conclusions of the PhD candidate's work that all of this is based on?

> **u/lovestruck90210** • 2 points • 2025-05-31 21:42 UTC
>
> literally in the post, in the section of my post that says "You can read more of De Vries-Gao's analysis [here](https://www.cell.com/joule/fulltext/S2542-4351(25)00142-4)"
>
>> **u/NuOfBelthasar** • 1 points • 2025-05-31 21:46 UTC
>>
>> My bad. I jumped straight into reading the white paper, read through it, and the first thing I saw after finishing was your insulting assumptions about the people here. So I skipped reading the rest of your post to reply.
>>
>> I guess I made unfair assumptions about you, too. I'm glad you bothered to read the primary source.